

06-06-2026

ML-4

- derivative •
- gradient descent
- practical: flow of learning

calculus → derivative → find slope
→ rate of change
→ direction of flow
→ slope of tangent line

A derivative measures the instantaneous rate of change of function.
↳ tells us exactly how fast output changes for a tiny change in input.

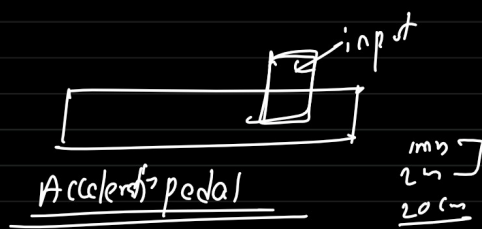
input → output
↳ tiny change

function → $3x^2 + 5$ → how much change?
output → $\uparrow$
input
derivative

input → x

→ $x + h$

$h \rightarrow 0.00001$
 $h \rightarrow -0.0001$



output $\rightarrow$ speed
 $\rightarrow$ 20km/h

1mm $\rightarrow$ 1mm

$$f_n \rightarrow 3x^2 + 5$$

$\rightarrow$ tiny nudge $\rightarrow$ input $\rightarrow x \rightarrow (x+h)$, $h \rightarrow$ very small amount

$$\rightarrow 3(x+h)^2 + 5$$

$\rightarrow$ using algebra $\rightarrow (a+b)^2 \rightarrow a^2 + b^2 + 2ab$

$$\rightarrow 3(x^2 + h^2 + 2xh) + 5$$

$$\rightarrow 3x^2 + 3h^2 + 3 \cdot 2xh + 5$$

$$\rightarrow 3x^2 + 3h^2 + 6xh + 5$$

$\rightarrow$ find total change

$$\text{change} = (\text{New output}) - (\text{original output})$$

$$= (\cancel{3x^2} + 3h^2 + 6xh + \cancel{5}) - (\cancel{3x^2} + \cancel{5})$$

$$= 6xh + 3h^2$$

→ calculate rate of change

$$\text{Rate} = \frac{6xh + 3h^2}{h} = \frac{\text{change in output}}{\text{change in input}}$$

Divide both terms by h :

$$= 6x + 3h$$

→ Nudge (h) → infinitely small → Derivative

$$h \rightarrow 0$$

$$6x + 3h \quad (h \rightarrow 0) \rightarrow 6x$$

$$3x^2 + 5 \rightarrow 6x$$

$$x=1 \rightarrow 6$$

$$x=2 \rightarrow 12$$

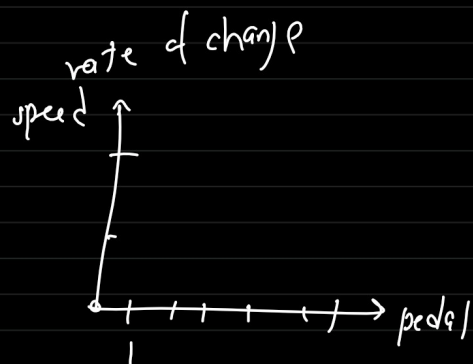
$$x=0 \rightarrow 0$$



rate of change at that moment

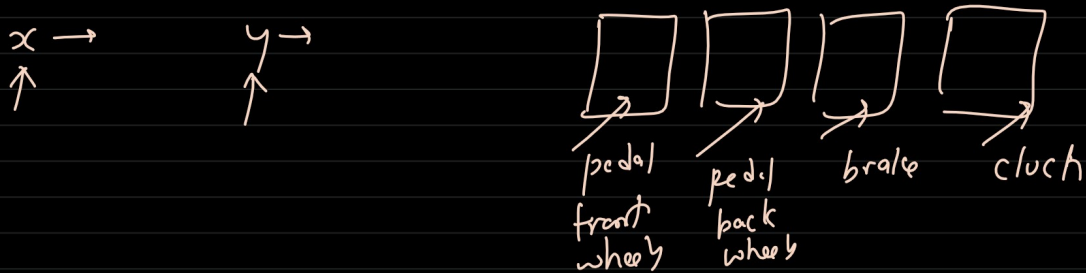
$$x=10$$

$$60$$



$3x^2 + 5$ ← original function
 $\uparrow$ ← one parameter

$3x^2 + 5xy + y^3$ ← 2 parameters (x, y)



multiple parameters → partial derivative

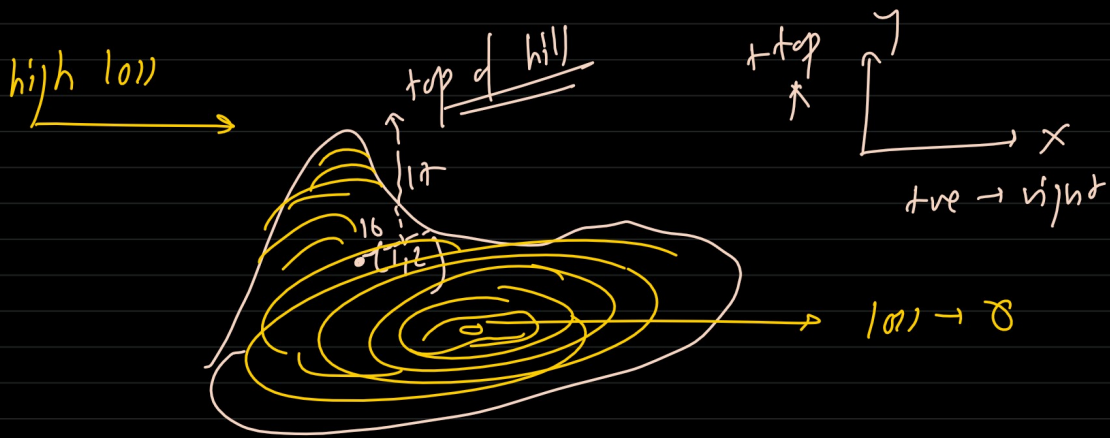
rate of change w.r.t → $\frac{df}{dx} \rightarrow 6x + 5y$

rate of change w.r.t → $\frac{df}{dy} \rightarrow 5x + 3y^2$

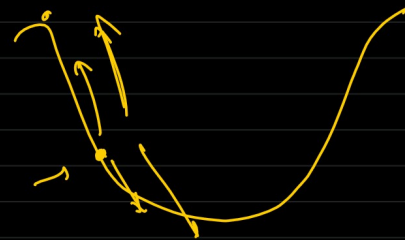
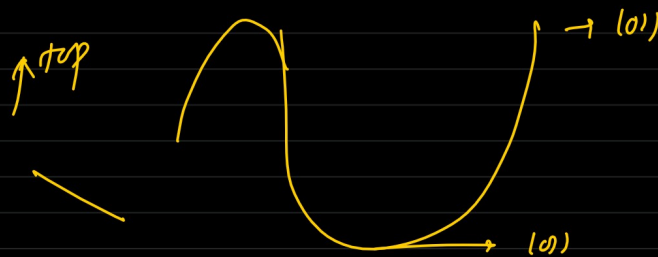
$$\nabla f(x, y) = (6x + 5y, 5x + 3y^2)$$

$$\nabla f(1, 2) = (6(1) + 5(2), 5(1) + 3(2)^2)$$

$\nabla f(1, 2) = \begin{pmatrix} 16 \\ \downarrow \text{right} \end{pmatrix}, \begin{pmatrix} 17 \\ \rightarrow \text{up} \end{pmatrix}$
 (gradient) derivation gives you direction of steepest ascent.



gradient always point towards the steepest ascent.



$$BMI = \frac{\text{weight}}{\text{height}^2} = \frac{w}{h^2}$$

parameters $\rightarrow w, h$
height $\rightarrow 1.8$

$\rightarrow$ change weight, keeping the height constant

$$\frac{df}{dw} = \frac{1}{h^2} = \frac{1}{(1.8)^2} \approx 0.309$$

$$w = 100$$

$$w = 101$$

$$\frac{df}{dh}$$

if my weight increases by 1 point/unit $\rightarrow$

$$BMI = BMI_{\text{old}} + \text{weight change} \times 0.309$$

$$2kj \rightarrow 2 \times 0.30 \rightarrow \underline{\underline{0.618}}$$

Gradient Descent

- goal is to find minimum loss
- we use differentiation to calculate slope of the cost function at our current position
- slope / gradient will tell us direction of steepest ascent (uphill hill)
- Take a step in the opposite direction → steepest descent (downhill)

Differentiation of the gradient:

$$\frac{1}{2m} (y - \hat{y})^2 \rightarrow \text{gradient}$$

(loss function) → modified MSE

$$\begin{array}{c} \text{error} \leftarrow y, \hat{y} \\ \uparrow \\ m_0 x_0, m_1 x_1, m_2 x_2, m_n x_n \\ \downarrow \downarrow \\ c \quad 1 \end{array}$$

$\beta \rightarrow \text{parameters} \rightarrow m_0, m_1, m_2 \rightarrow \hat{y}$
 $x \rightarrow \text{input features} \rightarrow$

← error → loss function

$J \rightarrow$ cost function, $\beta \rightarrow$ parameter (m_0, m_1, m_2)

$$\hat{y} = x_1 m_1 + x_2 m_2 + c$$

$$= x_1 m_1 + x_2 m_2 + x_0 m_0$$

Dataset:

Years of exp (x_1)	Completed projects (x_2)	Salary_k (y)
1	0	39.769
2	1	48.900
3	1	56.978
4	2	68.240
5	3	77.867
6	4	85.022

$$J(\beta) = \frac{1}{2n} (y - \hat{y})^2 \leftarrow \text{start point}$$

$$= \frac{1}{2n} (Y - X \cdot \beta)^2$$

$$q^2 \rightarrow a \times a$$

$$A^2 \rightarrow A \times A^T$$

$$= \frac{1}{2n} (Y - X \cdot \beta)^T (Y - X \cdot \beta)$$

constant

differentiation

keep it while

$$S(\beta) = (Y - X \cdot \beta)^T (Y - X \cdot \beta)$$

Rule 1: $(A - B)^T = A^T - B^T$

Rule 2: $(AB)^T = B^T \cdot A^T$

$$S(\beta) = (Y^T - (X\beta)^T) \cdot (Y - X \cdot \beta) \rightarrow \text{Rule 1}$$

$$= (Y^T - \underline{\beta^T X^T}) \cdot (Y - X \cdot \beta) \rightarrow \text{Rule 2}$$

$$= Y^T Y - Y^T X \cdot \beta - \beta^T X^T Y + \beta^T X^T X \beta$$

$$(\beta^T X^T Y) = (\beta^T X^T Y)^T$$

→ scalar rule

→ scalar is always equal to its own transpose

$$(\beta^T X^T Y)^T = Y^T \cdot X \cdot \beta$$

$$(ABC)^T = C^T B^T A^T$$

→ reversal property

$$= Y^T Y - Y^T X \beta - \beta^T X^T Y + \beta^T X^T X \beta$$

$$= Y^T Y - Y^T X \beta - Y^T X \beta + \beta^T X^T X \beta$$

$$= Y^T Y - 2 Y^T X \beta + \beta^T X^T X \beta$$

rate of change w.r.t β

Take the gradient: $\frac{\partial S}{\partial \beta}$, take partial derivative

$$\frac{\partial}{\partial \beta} (Y^T Y) = 0$$

$$\frac{\partial}{\partial x} (1) = 0$$

$$\frac{\partial}{\partial \beta} (-2 Y^T X \beta) = -2 (X^T Y)^T \beta = -2 X^T Y$$

$$\frac{\partial}{\partial x} (a^T x) = a$$

$$Y^T X = (X^T Y)^T$$

$$\frac{\partial}{\partial x} (x^T A x) = (A + A^T) x$$

$$\frac{\partial}{\partial \beta} \left(\underbrace{\beta^T}_{x^T} \underbrace{X^T}_A \underbrace{X \beta}_x \right) = \left(X^T X + \underbrace{(X^T X)^T}_{X^T \cdot (X^T)^T} \right) \beta$$

$X^T \cdot X$

$$= (X^T X + X^T X) \beta$$

$$= 2 X^T X \beta$$

combine:

$$\begin{aligned}\frac{\partial J}{\partial \beta} &= 0 - 2x^T y + 2x^T x \beta \\ &= 2x^T x \beta - 2x^T y \\ &= 2x^T (x\beta - y) \\ &= 2x^T (\hat{y} - y)\end{aligned}$$

bringing the constants:

$$J(\beta) = \frac{1}{2n} - 2x^T (\hat{y} - y)$$

$$= \frac{x^T (\hat{y} - y)}{n} \rightarrow \begin{array}{l} \text{total derivatives} \\ \text{of our cost} \\ \text{function} \\ \downarrow \\ \text{gradient} \end{array}$$

$$\text{gradient} = x.T @ (y_{pred} - y) / n$$

gradient
small = direction & rate of change $\times 0.01$
 $\uparrow$
learning rate

x (→ feature) + constant (1)

→ target column

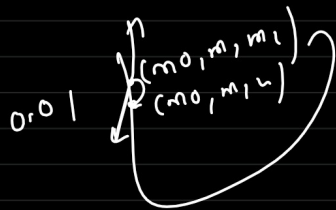
→ parameters (m_0, m_1, m_2)

→ loss function → error → cost function

→ gradient → direction of steepest ascent + rate of change

→ learning → control gradient / step

→ update the parameters (m_0, m_1, m_2)

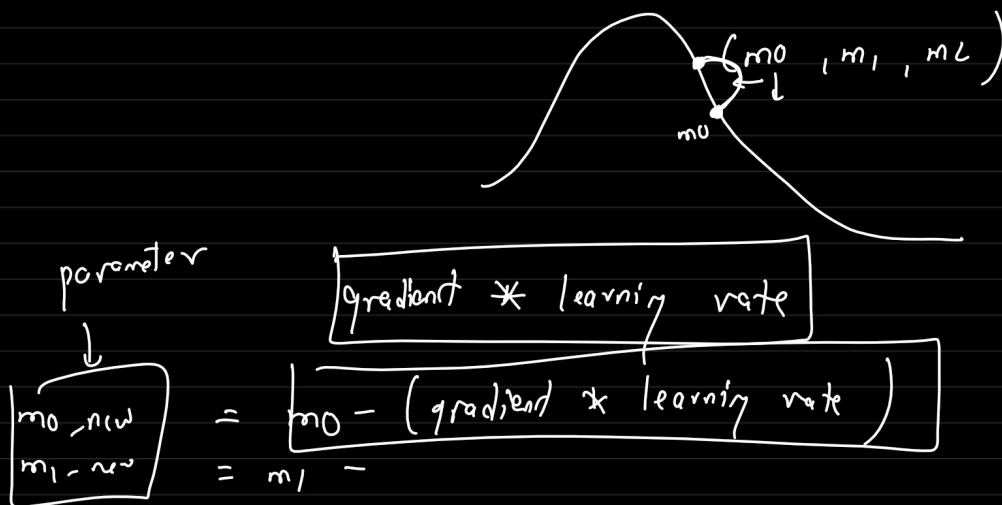


→ $\hat{y} = x \cdot \beta$

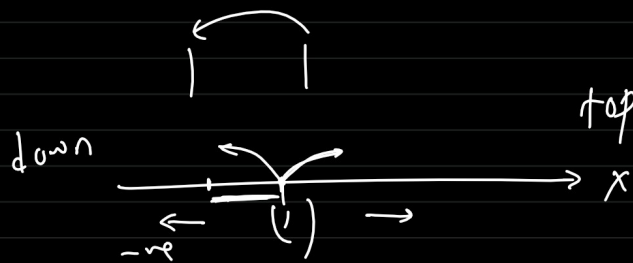
→ error = $y - \hat{y}$

→ gradient = → direction of steepest ascent + magnitude

→ gradient * learning

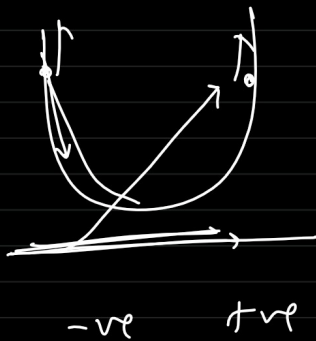


$$m_2 - \text{new} = m_2 -$$



$m_1 = 1$
gradient $\rightarrow$ direction + rate of change $\times$ learning rate

$$m_{1,\text{new}} = m_1 - \text{gradient} \times \text{lr}$$



left to right upwards $\rightarrow$ +ve slope
left to right downwards $\rightarrow$ -ve slope

$$y = x_1 m_1 + x_2 m_2 + x_3 m_3$$

1000 iterat $\rightarrow$ error

while (True)

1000 5 error)

1 - 10

$\rightarrow$ 11

$\rightarrow$ 9

$\rightarrow$ 8-9

- 10

linear regression:-

$\rightarrow$ Initialize weight to 0

for i in range (0, 1000):

$\rightarrow y_{\text{pred}} = x \cdot \beta$ } only for error

$\rightarrow$ print (error)

$\rightarrow \beta_{\text{new}} = \beta_{\text{old}} - \frac{\partial L}{\partial \beta} \times \text{learning}$

$\rightarrow$ gradient descent
 $\rightarrow$ actual parameter learning

linear regression

x_0	x_1	x_2
1	1	6
1	2	1
1	3	1

$m_0 \quad m_1 \quad m_2$

$$\rightarrow m_0 \cdot x_1 + m_1 \cdot x_1 + m_2 \cdot x_2$$

$$m_0 \rightarrow 0$$

$$m_1 \rightarrow 0$$

$$m_2 \rightarrow 0$$

$$\text{error} \rightarrow X \cdot M$$

$$\boxed{\begin{aligned} \hat{M} &= M - \text{gradient } X | \text{error} \\ \text{error} &= X \cdot M \end{aligned}}$$